

August 4, 2026

Dear Dr. Dardour and the Editorial Team of Pattern Recognition,

RE-SUBMISSION of Ms. Ref. No. PR-D-26-10696, previously titled "ATLAS: Anytime-valid Testing of Latent Simulators -- Sequential Certification of World-Model Faithfulness via E-Process Betting."

Thank you for your letter regarding Ms. Ref. No. PR-D-26-10696 and for the detailed formatting and desk-check comments. This is a re-submission of that manuscript, now retitled "ATLAS: Anytime-Valid Certification of World-Model Faithfulness via E-Process Betting" per Comment 1 below. I have thoroughly revised the manuscript to address every point raised and describe each change below; please retain this submission under the same manuscript reference, PR-D-26-10696, if your system allows, or treat it as the direct resubmission of that reference otherwise.

## POINT-BY-POINT RESPONSE

### Specific Comments

1. Title length and grammar. The title is now a single, succinct line: "ATLAS: Anytime-Valid Certification of World-Model Faithfulness via E-Process Betting" (9 words), replacing the previous two-line title-plus-subtitle.
2. Single-column, double-spaced, 10pt format via the official template. The manuscript is now typeset with `\documentclass[times, review, 10pt]{elsarticle}` exactly as specified, using the Elsevier `elsarticle` class (single column, "review" double line spacing, Times font, 10pt base size).
3. Equations. All equations remain native, editable LaTeX (no images) and now render at the document's 10pt base size. Several equations that overflowed the narrower single-column measure under the new template were reflowed into shorter displayed equations to fit cleanly within the text width.
4. Table font size and style. Both tables (Table 1, Table 2) are set at the document's 10pt base size (well above the 8pt minimum), use only horizontal booktabs rules (no vertical rules, no cell shading), and remain in standard portrait orientation.
5. Footnote, author name, affiliation, and caption font size. Footnotes and figure/table captions use LaTeX's `\footnotesize`, which resolves to exactly 8pt at the document's 10pt base size. The author name and affiliation block, which the class otherwise typesets at 10pt, has been explicitly set to 8pt to match.

6. Figure/table placement. All figures and tables are placed with standard floating placement close to their first in-text reference; I verified programmatically that every figure/table lands within 0-3 pages of the paragraph that first cites it, consistent with the surrounding figure-dense sections.

7. Reference count (35-55). The reference list now contains 39 entries, within the requested range for a regular paper.

8. Reference numbering by order of appearance. The bibliography style was switched from an author-year (Harvard) style to `elsarticle-num`, which numbers references [1], [2], ... in the order they are first cited in the text; I verified this ordering in the compiled PDF.

9. Page limit (20-35 pages, excluding Highlights, Author Biography, and Declaration of Interest). The revised manuscript is 34 pages including the reference list, within the limit. This despite substantial content growth (an expanded Discussion and conclusions section, two new citations with individual discussion), made possible by the more compact 10pt Times setting.

#### General Comments

(a) Title. Addressed under Specific Comment 1 above.

(b) Conclusions distinct from and longer than the abstract, covering strengths, weaknesses, benefit to the field, and future work. Section 6 ("Discussion and conclusions") has been substantially rewritten and expanded (from roughly 370 words to roughly 850 words, versus a 371-word abstract) into four explicit parts: Strengths (what the construction does not assume, representation-agnosticism, actionability, and the empirical delay/validity trade-off versus six published detectors); Weaknesses and limitations (four concrete limitations: unmeasured power under closed-loop policy feedback, the statistical cost of non-overlapping windows, dependence on world-model/representation quality, and the modeling choices embedded in the tolerance and scoring rule); How the field can build on this (three concrete adoption paths: a drop-in safety gate for practitioners, an audit template for conformal-monitoring researchers, and the generality of the betting construction to other sequentially-scored streams in pattern recognition); and Future work (closed-loop power, risk-calibrated tolerances, tightened interleaving, nonstationary-optimal betting, multi-modal world models, and hardware-in-the-loop validation).

(c) Bibliography currency, diversity, and avoiding uncommented citation groups. I reviewed every citation group in Related Work; the largest group is three references, and each group is already attached to a named method or result rather than a bare list. I added two individually discussed 2024-2025 citations from Pattern Recognition itself (see (d) below) and confirmed the bibliography is not concentrated in a single conference series or dominated by arXiv-only preprints (9 of 39 references, or 23%, are arXiv-only; the remainder are published journal or

conference papers spanning NeurIPS, CVPR, SDM, ICPR, ICML, RLC, and statistics journals including the Annals of Statistics, JRSS-A/B, and JMLR).

(d) Relevance to the Pattern Recognition readership. I added and individually discussed two recent papers published in Pattern Recognition itself: Msaddi and Kumbasar (2025), "Expanding conformal prediction to system identification" (Pattern Recognition), which extends conformal prediction to dynamical-system identification and is contrasted with ATLAS's online, deployment-time rollout certificate in the Related Work section; and Su, Tan, An, Xing, and Feng (2024), "Semantic-driven dual consistency learning for weakly supervised video anomaly detection" (Pattern Recognition, vol. 157, art. 110898), which is contrasted with ATLAS as an offline, frame-level anomaly localizer rather than a time-uniform faithfulness certificate for a generative model's own rollouts.

(e) Page and format limits. Addressed under Specific Comments 2 and 9 above; the manuscript is single-column, double-spaced, and 34 pages.

#### ADDITIONAL SUBMISSION INFORMATION (unchanged from the original submission)

Importance and fit to scope. World models -- learned, action-conditioned simulators used for video prediction, autonomous driving, and model-based decision making -- are now central to pattern recognition research, yet the field still evaluates them with a single offline number (Frechet Video Distance, reconstruction error, return gap) computed once before deployment. This manuscript shows that such offline scores are structurally blind to how a world model's faithfulness actually behaves: it decays with rollout horizon and drifts over deployment time, especially once the model is embedded in a closed decision loop that endogenously shapes its own test data. ATLAS instead gives a deployed world model a running, anytime-valid statistical certificate -- a trust horizon that a downstream planner can use directly to clip its imagination depth -- validated on a controllable synthetic simulator, four real video benchmarks (Moving MNIST, KTH Actions, KITTI driving, and UZH-FPV drone flight), and learned neural world models.

Comparison to the state of the art. The manuscript positions ATLAS against, and empirically compares it with, recent representative work, including Shin, Ramdas, and Rinaldo (2024, "E-detectors"), the closest anytime-valid change-detection framework, compared against directly in Table 2; Gibbs and Candes (2021) and its 2025-2026 online-conformal descendants (Leong, 2026; Opoku and Banahene, 2026), representative of the conformal-prediction baselines in Table 2; Assran et al. (2025, "V-JEPA 2") and NVIDIA et al. (2025, "Cosmos"), state-of-the-art world models exemplifying the class of representation ATLAS certifies; and, newly added in this revision, Msaddi and Kumbasar (2025) and Su et al. (2024) from Pattern Recognition itself. We are not aware of prior work that gives a horizon-resolved, anytime-valid faithfulness certificate for world models under policy-endogenous data.

Relationship to prior publications. This manuscript is not an extension of a previously published conference paper; it is original, full-length work.

AI-assistance disclosure. Generative AI (Claude, via Claude Code) was used to help draft, edit, and polish portions of the manuscript text, as disclosed in the manuscript itself. I reviewed all AI-assisted content and take full responsibility for the paper.

Thank you for the opportunity to revise the manuscript. I believe it now fully complies with the journal's formatting requirements and have strengthened its content and positioning accordingly. I would be glad to suggest reviewers or provide any further information the editorial process requires.

Sincerely,

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